D6 Executive summary

When someone lists a home for sale, one of the biggest questions is whether the asking price is actually realistic. My housing project's goal was to answer two practical questions using real housing sales data from Ames, Iowa. First, how much is a home likely to sell for based on its characteristics? Second, can we identify whether a property belongs to the "premium" segment of the market, meaning the top 25% of homes by price?

To answer these questions, I used the Ames Housing dataset containing nearly 3,000 residential property sales between 2006 and 2010. Each home included information such as living area size, number of bathrooms, garage capacity, neighborhood, year built, and overall quality rating. The project started in Jupyter Notebook, where I worked through data cleaning, exploratory data analysis, preprocessing, and feature engineering before training the machine learning models.

Several models were compared, ranging from simpler linear models to more advanced XGBoost models. One thing I focused on throughout the project was making sure the evaluation process stayed realistic and leakage-safe, so the results would reflect how the model would actually behave on unseen homes rather than just looking impressive on paper.

The final results were surprisingly strong. The regression model was typically able to predict sale prices within roughly $12,000 of the actual selling price, while the classification model identified premium homes with around 96% accuracy.

One of the most interesting findings for me was that the biggest drivers of price were also the factors people intuitively expect to matter most: overall quality, living area size, neighborhood, garage capacity, and the age of the property. I also found it interesting that simpler and easier-to-explain models performed almost as well as the more advanced ones. In my opinion, that is important because in real-world situations people usually want tools they can actually understand and trust, not only highly complex systems that are difficult to explain.

At the same time, this project also showed me that machine learning should be viewed as a "teammate" rather than a replacement for human judgment. In my opinion, tools like this work best as a second opinion alongside critical judgment and market expertise, especially because the model is based on historical Ames housing data and performs best on homes similar to those seen during training.

In my opinion, one of the biggest takeaways from the project was realizing how important careful verification and testing really are. Even when models produce strong results, it is still necessary to check outputs carefully and make sure the system behaves realistically outside the notebook environment.

The final outcome of the project was a fully deployed interactive Streamlit web application where users can enter home characteristics, receive an estimated sale price, see whether the home is classified as premium or if it has an investment potential and explore interactive housing market insights based on the Ames dataset.